Precision Vehicle Tracking and Disturbance Recovery

Reference-Scheduled LPV-MPC

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Abstract

A reference-scheduled linear parameter-varying model predictive controller (LPV-MPC) is formulated for vehicle trajectory tracking and disturbance recovery. A nonlinear kinematic bicycle model is expressed in moving-frame error coordinates and linearized along a known reference. A sparse quadratic program optimizes steering and acceleration over a two-second horizon while enforcing command, command-rate and predicted-speed limits. Evaluation on a variable-speed oval, a double-loop trajectory and a disturbed lane-change maneuver produced 1,520 successful optimization solves and full-run position root-mean-square errors of 0.289–0.396 m. Following removal of the lane-change disturbance, cross-track and speed errors met the specified settling thresholds within 1.3 s. The results demonstrate tracking and finite-run recovery under the evaluated kinematic simulation conditions and identify the modeling and validation requirements for broader application.

1 Introduction

Vehicle trajectory tracking requires a controller to regulate position, heading and speed while respecting actuator limits. Curved paths, changing reference speeds and disturbances couple these objectives: aggressive corrections can reduce error but also produce abrupt steering or acceleration. Model predictive control addresses this trade-off by optimizing a sequence of commands over a finite horizon and applying the first command before updating the prediction from the measured state [1].

This study investigates reference-scheduled linear parameter-varying model predictive control (LPV-MPC) for constrained vehicle tracking and disturbance recovery. A nonlinear kinematic bicycle model is expressed in a moving reference frame, and its local prediction matrices vary with reference speed and curvature. The controller uses this preview to optimize steering and acceleration subject to command, command-rate and speed bounds.

The evaluation considers a variable-speed oval, a double-loop trajectory and a lane-change maneuver. Each run begins with position, heading and speed errors; the lane-change experiment also includes a temporary lateral drift and longitudinal load absent from the prediction model. Tracking error, constraint satisfaction, optimization outcomes and post-disturbance recovery quantify performance. The following sections present the model and controller formulation, numerical results, and their interpretation.

2 Concepts and Methods

The controller combines a moving-frame tracking model, reference-dependent linear prediction and a constrained quadratic program. The plant is propagated separately using nonlinear dynamics.

2.1 Notation, assumptions, and system architecture

Symbol	Meaning	Unit
$\boldsymbol{z} = [x, y, \psi, v]^\top$	Vehicle position, heading, and speed	m, rad, m/s
$\boldsymbol{u} = [\delta, a]^\top$	Front-wheel steering and acceleration command	rad, m/s^2
L	Wheelbase, fixed at 2.7	m
$(\cdot)_r$	Reference quantity at the same time as the state	varies
κ_r, ω_r	Reference curvature and yaw rate	m^{-1}, rad/s
$\boldsymbol{e} = [e_s, e_y, e_\psi, e_v]^\top$	Along-track, cross-track, heading, speed errors	varies
T_s, N	Sample period 0.1 s; prediction horizon 20	s, steps
$\boldsymbol{e}_{i\|k}, \boldsymbol{u}_{i\|k}$	Predictions i steps ahead, computed at sample k	varies
d_y, d_a	Global lateral drift and longitudinal load	m/s, m/s^2
Q, R, S, P_f	Error, feedforward, rate, terminal weights	scaled weights
$\boldsymbol{\xi}$	Stacked QP decision vector	mixed

Assumptions. The reference is analytic and available beyond the end of each run, so every optimization receives a complete preview. Reference speed is nonzero. The controller receives the exact simulated state; sensor noise, estimation errors, latency, and actuator lag are absent. Steering and acceleration commands are held constant during each control interval. The plant is planar and kinematic, with no tire-slip dynamics.

The feedback architecture is shown in Figure 1.

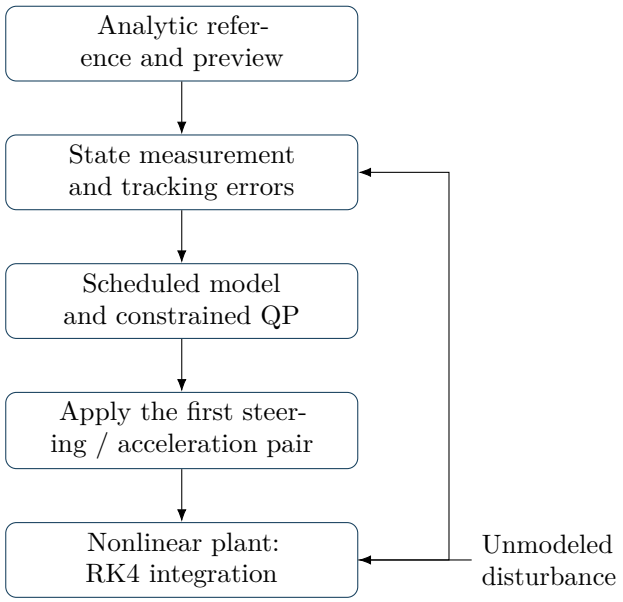


Figure 1: Receding-horizon LPV-MPC control architecture.

Time-indexed tracking differs from geometric path following. A vehicle can lie on the correct curve yet have nonzero along-track error because it arrives too early or too late. The objective here penalizes that timing-related error.

2.2 Nonlinear vehicle and feasible reference motion

2.2.1 Kinematic bicycle plant

The nominal model maps vehicle speed and steering to planar translation and rotation:

$$\dot{\boldsymbol{z}} = \begin{bmatrix} v\cos\psi \\ v\sin\psi + d_y(t) \\ (v/L)\tan\delta \\ a + d_a(t) \end{bmatrix}. \tag{2.1}$$

The yaw-rate relation follows from a turning radius $R_c = L/\tan\delta$, giving $\dot{\psi} = v/R_c$. It assumes no lateral slip at the modeled reference point. The injected d_y is an additive drift in the global y direction; it is not a tire-force model. Likewise, d_a is an additive acceleration disturbance, not an identified powertrain load.

2.2.2 Reference geometry and feedforward commands

Let $\boldsymbol{p}_r(t) = [x_r(t), y_r(t)]^\top$ be twice differentiable with $v_r > 0$. The implementation calculates

$$v_r = \sqrt{\dot{x}_r^2 + \dot{y}_r^2}, \qquad \psi_r = \operatorname{atan2}(\dot{y}_r, \dot{x}_r), \tag{2.2}$$

$$\kappa_r = \frac{\dot{x}_r\ddot{y}_r - \dot{y}_r\ddot{x}_r}{v_r^3}, \qquad a_r = \frac{\dot{x}_r\ddot{x}_r + \dot{y}_r\ddot{y}_r}{v_r}, \tag{2.3}$$

$$\omega_r = v_r\kappa_r, \qquad \delta_r = \arctan(L\kappa_r). \tag{2.4}$$

Thus $\boldsymbol{u}_r = [\delta_r, a_r]^\top$ is the command that reproduces the reference in the nominal continuous model. Substitution gives $\dot{\psi}_r = v_r\tan\delta_r/L = \omega_r$ and $\dot{v}_r = a_r$. These identities ensure that zero tracking error with the feedforward command is a consistent nominal motion.

2.2.3 Numerical propagation

The controller updates at $T_s = 0.1$ s. The plant uses five RK4 substeps per interval, each of length $h = 0.02$ s. With a held command and derivative f from (2.1), one substep is

$$\boldsymbol{k}_1 = f(\boldsymbol{z}, t), \qquad \boldsymbol{k}_2 = f(\boldsymbol{z} + \tfrac{h}{2}\boldsymbol{k}_1, t + \tfrac{h}{2}), \tag{2.5}$$

$$\boldsymbol{k}_3 = f(\boldsymbol{z} + \tfrac{h}{2}\boldsymbol{k}_2, t + \tfrac{h}{2}), \qquad \boldsymbol{k}_4 = f(\boldsymbol{z} + h\boldsymbol{k}_3, (t+h)^-), \tag{2.6}$$

$$\boldsymbol{z}^+ = \boldsymbol{z} + \frac{h}{6}(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4). \tag{2.7}$$

The endpoint is evaluated from the left at scheduled disturbance boundaries. Commands are applied without clipping, allowing the recorded inputs to test the optimizer's constraints directly. Heading is wrapped after propagation.

2.3 Derivation of moving-frame tracking errors

Define a rotation from the global frame to the reference tangent-normal frame:

$$T(\psi_r) = \begin{bmatrix} \cos\psi_r & \sin\psi_r \\ -\sin\psi_r & \cos\psi_r \end{bmatrix}, \qquad \begin{bmatrix} e_s \\ e_y \end{bmatrix} = T(\psi_r)(\boldsymbol{p} - \boldsymbol{p}_r). \tag{2.8}$$

Positive e_s means the vehicle is ahead of the reference along its tangent; positive e_y means it lies to the reference's left. Define

$$e_\psi = \mathrm{wrap}(\psi - \psi_r), \qquad e_v = v - v_r. \tag{2.9}$$

The angle wrapping selects the local error in $[-\pi, \pi]$; the following differential equations apply away from the branch discontinuity.

2.3.1 Position-error dynamics

Writing $\boldsymbol{e}_p = [e_s, e_y]^\top$ and differentiating the rotating-frame expression yields

$$\dot{\boldsymbol{e}}_p = \dot{T}T^\top \boldsymbol{e}_p + T(\dot{\boldsymbol{p}} - \dot{\boldsymbol{p}}_r), \qquad \dot{T}T^\top = \begin{bmatrix} 0 & \omega_r \\ -\omega_r & 0 \end{bmatrix}. \tag{2.10}$$

The nominal vehicle velocity in this frame is $[v\cos e_\psi, v\sin e_\psi]^\top$, while reference velocity is $[v_r, 0]^\top$. Therefore the exact nominal error dynamics are

$$\dot{\boldsymbol{e}} = \begin{bmatrix} (v_r + e_v)\cos e_\psi - v_r + \omega_r e_y \\ (v_r + e_v)\sin e_\psi - \omega_r e_s \\ \dfrac{v_r + e_v}{L}\tan\delta - \omega_r \\ a - a_r \end{bmatrix}. \tag{2.11}$$

The terms $+\omega_r e_y$ and $-\omega_r e_s$ arise from frame rotation. Omitting them would change the model on curved trajectories. The position errors are not computed by searching for the nearest waypoint: the frame is anchored at the reference time.

2.3.2 How the disturbance enters these coordinates

The global disturbance transforms as $T[0, d_y]^\top$. The disturbed error model adds

$$\boldsymbol{w}_e(t) = \begin{bmatrix} \sin\psi_r\, d_y \\ \cos\psi_r\, d_y \\ 0 \\ d_a \end{bmatrix} \tag{2.12}$$

to (2.11). This expression explains why a global lateral disturbance need not act exclusively on cross-track error during a turn. The controller omits $\boldsymbol{w}_e$ and reacts to its effects through subsequent measured errors.

2.4 Reference-scheduled linear prediction model

2.4.1 Linearization about a reference motion

Linearize (2.11) at $\boldsymbol{e} = 0$ and $\boldsymbol{u} = \boldsymbol{u}_r$. With $\widetilde{\boldsymbol{u}} = \boldsymbol{u} - \boldsymbol{u}_r$, the local approximation is

$$\dot{\boldsymbol{e}} \simeq A_c(\rho)\boldsymbol{e} + B_c(\rho)\widetilde{\boldsymbol{u}}, \qquad \rho = [v_r, \kappa_r]^\top. \tag{2.13}$$

Since the nominal derivative vanishes at the expansion point, there is no independent affine term in these error coordinates. The Jacobians are

$$A_c = \begin{bmatrix} 0 & \omega_r & 0 & 1 \\ -\omega_r & 0 & v_r & 0 \\ 0 & 0 & 0 & \kappa_r \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{v_r}{L\cos^2\delta_r} & 0 \\ 0 & 1 \end{bmatrix}. \tag{2.14}$$

For example, the heading-error derivative with respect to speed error is $\tan\delta_r/L = \kappa_r$, and its derivative with respect to steering is $v_r \sec^2\delta_r/L$. These are distinct couplings: speed changes affect heading on a curved reference even when steering remains at its feedforward value.

2.4.2 Discretization and preview scheduling

Forward Euler yields

$$A_i = I + T_s A_c(\rho_{i|k}), \qquad B_i = T_s B_c(\rho_{i|k}), \tag{2.15}$$

$$\boldsymbol{e}_{i+1|k} = A_i \boldsymbol{e}_{i|k} + B_i(\boldsymbol{u}_{i|k} - \boldsymbol{u}_{r,i|k}), \quad i = 0, \ldots, N-1. \tag{2.16}$$

The horizon is $NT_s = 2.0$ s. Each pair (A_i, B_i) is calculated from the corresponding future reference sample. The scheduling sequence is known before optimization; predicted states do not update it inside a nonlinear iteration.

2.4.3 Meaning of the LPV description

The model is linear in errors and command deviations, with coefficients depending on reference parameters. For a fixed preview it is a linear time-varying (LTV) prediction sequence. It is a *local reference-scheduled LPV approximation*, not an exact global LPV representation of the nonlinear vehicle. Euler discretization and neglected higher-order terms contribute prediction error. Separate RK4 plant integration ensures the evaluation does not merely propagate the same linear model used by the optimizer.

2.5 Physical limits and horizon constraints

The decision variables are absolute steering and acceleration commands. Their amplitude bounds are

$$\boldsymbol{u}_{\min} \le \boldsymbol{u}_{i|k} \le \boldsymbol{u}_{\max}, \quad \boldsymbol{u}_{\min} = \begin{bmatrix} -25^\circ \\ -2.5 \end{bmatrix}, \quad \boldsymbol{u}_{\max} = \begin{bmatrix} 25^\circ \\ 2.0 \end{bmatrix}. \tag{2.17}$$

All angular quantities are converted to radians in the implementation. Acceleration is in $\mathrm{m/s^2}$.

2.5.1 Rate limits and the previously applied command

Define $\Delta\boldsymbol{u}_{0|k} = \boldsymbol{u}_{0|k} - \boldsymbol{u}_{k-1}$ and $\Delta\boldsymbol{u}_{i|k} = \boldsymbol{u}_{i|k} - \boldsymbol{u}_{i-1|k}$ for $i \ge 1$. The inequalities are

$$-\boldsymbol{r}T_s \le \Delta\boldsymbol{u}_{i|k} \le \boldsymbol{r}T_s, \qquad \boldsymbol{r} = \begin{bmatrix} 35^\circ/\mathrm{s} \\ 3.0\ \mathrm{m/s^3} \end{bmatrix}. \tag{2.18}$$

The acceleration-rate constraint bounds *commanded jerk*. It does not bound the derivative of total physical acceleration when an external load changes abruptly. At the first sample $\boldsymbol{u}_{-1} = 0$, limiting initial changes to 3.5° steering and $0.3\ \mathrm{m/s^2}$ acceleration per interval. This prevents the controller from instantly applying a large correction to the initial error.

2.5.2 Predicted-speed bounds

Because $v_{i|k} = v_{r,i|k} + e_{v,i|k}$, the limits are linear in the error state:

$$1 - v_{r,i|k} \le e_{v,i|k} \le 10 - v_{r,i|k}, \qquad i = 1, \dots, N. \tag{2.19}$$

These apply to predicted future samples. The initial measured error is imposed as an equality; speed at sample zero is not an additional QP bound. The saved nonlinear plant trajectory is checked separately for sampled speed violations.

2.5.3 Why feasibility matters

All constraints are hard; there are no slack variables. A previously applied command far outside the amplitude range can make it impossible to re-enter that range within one allowed rate step. For instance, a prior steering of 2 rad cannot reach $25^\circ \simeq 0.4363$ rad with a maximum step of approximately 0.0611 rad. The numerical test suite deliberately constructs this infeasible case.

Constraint satisfaction for the predicted model does not automatically establish constraint satisfaction between samples or under arbitrary plant mismatch. The current evaluated runs include independent sampled checks; no collision or road-boundary constraints are present.

2.6 Objective function and tuning interpretation

At sample k, the finite-horizon objective is

$$\begin{aligned} J_k = & \sum_{i=0}^{N-1} \boldsymbol{e}_{i|k}^\top Q \boldsymbol{e}_{i|k} + \boldsymbol{e}_{N|k}^\top P_f \boldsymbol{e}_{N|k} \\ & + \sum_{i=0}^{N-1} (\boldsymbol{u}_{i|k} - \boldsymbol{u}_{r,i|k})^\top R (\boldsymbol{u}_{i|k} - \boldsymbol{u}_{r,i|k}) \\ & + \sum_{i=0}^{N-1} \Delta \boldsymbol{u}_{i|k}^\top S \Delta \boldsymbol{u}_{i|k}. \end{aligned} \tag{2.20}$$

The selected matrices are

$$Q = \mathrm{diag}(4, 22, 12, 4), \quad R = \mathrm{diag}(3, 0.25), \quad S = \mathrm{diag}(12, 0.8), \quad P_f = 3Q. \tag{2.21}$$

The implemented state weights act on meters, radians, and m/s. Their numerical sizes should not be compared as if all errors had the same units or scale.

2.6.1 What each term does

- **Tracking error.** The Q term rewards agreement with the reference motion. Cross-track error is emphasized, while along-track and speed error remain explicit objectives.
- **Feedforward deviation.** The R term penalizes departure from the nominal steering and acceleration required by the reference. On a curve, correct steering is generally nonzero; penalizing deviation from δ_r avoids treating all steering as unnecessary effort.
- **Command smoothness.** The S term discourages rapid command changes even when they remain inside the hard rate limits. Rate bounds define admissibility; the rate penalty ranks admissible sequences.

- **Terminal tracking.** $P_f = 3Q$ adds weight to the final error. It is a design choice, not a Riccati-derived terminal certificate or an invariant terminal-set construction.

2.6.2 Interpretation of trade-offs

Increasing tracking weights tends to favor faster correction, potentially driving commands toward their bounds. Increasing rate penalties tends to smooth commands, potentially extending recovery. The actual trade-off depends on state units, reference curvature, sampling, constraints, and horizon length. No tuning sweep was performed for this report; these are interpretations of the objective, not measured comparisons between alternative settings.

The current measured error $\boldsymbol{e}_{0|k}$ is fixed, so its quadratic term contributes a constant to the optimization. Retaining it in the stacked objective simplifies the formulation without changing the minimizing command sequence.

2.7 Sparse quadratic-program construction

Define the stacked error and command vectors

$$\boldsymbol{E} = \begin{bmatrix} \boldsymbol{e}_{0|k} \\ \vdots \\ \boldsymbol{e}_{N|k} \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} \boldsymbol{u}_{0|k} \\ \vdots \\ \boldsymbol{u}_{N-1|k} \end{bmatrix}, \quad \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{E} \\ \boldsymbol{U} \end{bmatrix} \in \mathbb{R}^{124}. \tag{2.22}$$

There are $4(N+1) = 84$ error-state entries and $2N = 40$ command entries. Define

$$\bar{Q} = \operatorname{blkdiag}(Q, \ldots, Q, P_f), \quad \bar{R} = I_N \otimes R, \quad \bar{S} = I_N \otimes S, \tag{2.23}$$

$$D = \begin{bmatrix} I_2 & 0 & \cdots & 0 \\ -I_2 & I_2 & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & -I_2 & I_2 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} \boldsymbol{u}_{k-1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{2.24}$$

Thus $D\boldsymbol{U} - \boldsymbol{b}$ is the stacked command-increment vector. Expanding (2.20) and dropping constants gives the solver form

$$\min_{\boldsymbol{\xi}} \ \tfrac{1}{2}\boldsymbol{\xi}^\top H \boldsymbol{\xi} + \boldsymbol{q}^\top \boldsymbol{\xi} \quad \text{subject to} \quad \boldsymbol{\ell} \le C\boldsymbol{\xi} \le \boldsymbol{h}, \tag{2.25}$$

where

$$H = 2\operatorname{blkdiag}(\bar{Q}, \bar{R} + D^\top \bar{S} D), \quad \boldsymbol{q} = \begin{bmatrix} 0 \\ -2(\bar{R}\boldsymbol{U}_r + D^\top \bar{S}\boldsymbol{b}) \end{bmatrix}. \tag{2.26}$$

The factor of two follows from the solver's $\frac{1}{2}\boldsymbol{\xi}^\top H \boldsymbol{\xi}$ convention. Omitting it from only one objective component would change the relative weights.

2.7.1 Dynamics and inequality blocks

The first four constraint rows impose $\boldsymbol{e}_{0|k} = \boldsymbol{e}_k$. For each step, the equality row is

$$-A_i \boldsymbol{e}_{i|k} + \boldsymbol{e}_{i+1|k} - B_i \boldsymbol{u}_{i|k} = -B_i \boldsymbol{u}_{r,i|k}. \tag{2.27}$$

Equality rows use identical lower and upper bounds. The remaining blocks select $\boldsymbol{U}$, $D\boldsymbol{U}$, and future speed errors. In total, $C \in \mathbb{R}^{184\times 124}$ has 84 equality rows, 40 amplitude rows, 40 rate rows, and 20 speed rows. Keeping states explicit produces a sparse dynamics structure rather than eliminating them into a dense command-only problem.

2.8 Optimization, feedback, and numerical acceptance

2.8.1 Receding-horizon procedure

1. At t_k, calculate $\boldsymbol{e}_k$ from the measured vehicle state and reference.
2. Obtain $N+1$ reference samples; construct N scheduled model pairs and nominal commands.
3. Assemble $\boldsymbol{q}$, C, $\boldsymbol{\ell}$, and $\boldsymbol{h}$, incorporating the previously applied command.
4. Solve (2.25), check solver status, finite values, and constraint residuals.
5. Apply only $\boldsymbol{u}^{\star}_{0|k}$ for one interval; integrate the nonlinear plant with RK4.
6. Measure the new state and repeat with a shifted reference window.

Feedback comes from repeatedly replacing the initial predicted error by the measured error. A disturbance creates a discrepancy between the previous prediction and the next measurement, which triggers a new corrective solution.

2.8.2 Numerical solver

OSQP solves the convex QP using an operator-splitting method based on the alternating direction method of multipliers (ADMM) [2, 3]. The implementation requests absolute and relative tolerances of 10^{-6}, permits 12,000 iterations, and enables polishing. Only the `solved` status is accepted. A fresh solver is created at every control step; the previous primal solution is supplied as a warm start. The implementation does not reuse a persistent factorization across steps or explicitly shift the previous trajectory before warm starting.

An independent feasibility check computes

$$\epsilon_c = \max\left(0, \max_j[\ell_j - (C\boldsymbol{\xi})_j], \max_j[(C\boldsymbol{\xi})_j - h_j]\right). \tag{2.28}$$

A solution is rejected if any entry is nonfinite or $\epsilon_c > 2\times 10^{-5}$. This mixed-unit maximum is an implementation acceptance threshold, not a normalized physical safety margin. Failure raises an error; a fallback policy is not implemented.

2.8.3 Convexity and solution uniqueness

Every diagonal entry of Q, P_f, and R is positive, while S is positive definite. Hence $\bar{R} + D^{\top}\bar{S}D \succ 0$ and $H \succ 0$. All constraints are affine for a fixed reference preview. Consequently, a feasible finite QP has a unique minimizing decision vector. This is a property of the prediction problem; it does not establish global optimality of the nonlinear closed-loop motion.

2.9 Worked prediction model

Consider a reference point with $v_r = 4$ m/s, $\kappa_r = 0.02$ m^{-1}, and $L = 2.7$ m. Then

$$\omega_r = 0.08 \text{ rad/s}, \qquad \delta_r = \arctan(0.054) \simeq 0.05395 \text{ rad}. \tag{2.29}$$

Since $\sec^2(\arctan q) = 1 + q^2$, the steering gain is

$$g_\delta = \frac{4}{2.7}(1 + 0.054^2) \simeq 1.48580 \text{ s}^{-1}. \tag{2.30}$$

At $T_s = 0.1$ s, the prediction matrices become

$$A_d = \begin{bmatrix} 1 & 0.008 & 0 & 0.1 \\ -0.008 & 1 & 0.4 & 0 \\ 0 & 0 & 1 & 0.002 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.14858 & 0 \\ 0 & 0.1 \end{bmatrix}. \tag{2.31}$$

For $\boldsymbol{e} = [-1, 1.4, 0.139626, -0.7]^\top$ and zero command deviation from feedforward,

$$\boldsymbol{e}^+ \simeq A_d \boldsymbol{e} = [-1.0588,\ 1.46385,\ 0.138226,\ -0.7]^\top. \tag{2.32}$$

Thus nominal feedforward alone does not remove the offset. The positive cross-track increment is consistent with the heading error and frame-rotation terms. A feedback optimization must choose command deviations to alter this evolution.

2.9.1 Horizon structure

For $N = 20$, the decision vector contains 21 four-dimensional error states and 20 two-dimensional commands. A single dynamics equality couples only $\boldsymbol{e}_i$, $\boldsymbol{e}_{i+1}$, and $\boldsymbol{u}_i$. The difference operator couples adjacent commands. This block-local structure explains the sparse assembly.

With only two predicted intervals, the dynamics rows would have the form

$$\begin{bmatrix} I & 0 & 0 & 0 & 0 \\ -A_0 & I & 0 & -B_0 & 0 \\ 0 & -A_1 & I & 0 & -B_1 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_0 \\ \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \\ \boldsymbol{u}_0 \\ \boldsymbol{u}_1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{e}_k \\ -B_0 \boldsymbol{u}_{r,0} \\ -B_1 \boldsymbol{u}_{r,1} \end{bmatrix}. \tag{2.33}$$

The full horizon repeats this pattern. Absolute commands appear in the decision vector, while reference commands appear on the right-hand side and in the objective's linear term.

2.10 Optimality conditions

Rewrite (2.25) with equality constraints $A_{\text{eq}}\boldsymbol{\xi} = \boldsymbol{b}_{\text{eq}}$ and inequalities $G\boldsymbol{\xi} \le \boldsymbol{g}$. For equality multipliers $\boldsymbol{\lambda}$ and nonnegative inequality multipliers $\boldsymbol{\mu}$, the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\boldsymbol{\xi}^\top H \boldsymbol{\xi} + \boldsymbol{q}^\top \boldsymbol{\xi} + \boldsymbol{\lambda}^\top (A_{\text{eq}}\boldsymbol{\xi} - \boldsymbol{b}_{\text{eq}}) + \boldsymbol{\mu}^\top (G\boldsymbol{\xi} - \boldsymbol{g}). \tag{2.34}$$

The Karush–Kuhn–Tucker conditions are

$$H\boldsymbol{\xi}^\star + \boldsymbol{q} + A_{\text{eq}}^\top \boldsymbol{\lambda}^\star + G^\top \boldsymbol{\mu}^\star = 0, \tag{2.35}$$

$$A_{\text{eq}}\boldsymbol{\xi}^\star = \boldsymbol{b}_{\text{eq}}, \tag{2.36}$$

$$G\boldsymbol{\xi}^\star \le \boldsymbol{g}, \qquad \boldsymbol{\mu}^\star \ge 0, \tag{2.37}$$

$$\mu_j^\star (G\boldsymbol{\xi}^\star - \boldsymbol{g})_j = 0 \quad \text{for every } j. \tag{2.38}$$

These conditions express a balance between the objective gradient and the active constraints. When a rate limit is active, the optimizer cannot improve tracking simply by increasing that command change. It must redistribute the correction across later steps and possibly other commands. This provides a theoretical interpretation of the saturated rate traces.

For this strictly convex QP with affine constraints, a feasible optimum is unique, and multipliers satisfying these conditions certify its optimality. Numerical convergence and feasibility are evaluated to tolerance. The controller does not store dual histories in the published result arrays, so the report does not infer multiplier values or quantify constraint shadow prices.

2.11 Experimental design and performance measures

2.11.1 Analytic reference trajectories

Three test cases are considered. Case I is a variable-speed elliptical trajectory. Case II is a double-loop trajectory shaped like the numeral 8, with a central crossing and changes in turn direction. Case III is a lane-change trajectory with a temporary disturbance. Their reference coordinates are defined directly in time:

$$\text{Case I:} \quad x_r = 42\cos\theta, \quad y_r = 25\sin\theta, \quad \theta = 2\pi t/52, \tag{2.39}$$

$$\text{Case II:} \quad x_r = 45\sin\theta, \quad y_r = 16\sin 2\theta, \quad \theta = 2\pi t/64, \tag{2.40}$$

$$\text{Case III:} \quad x_r = 5t, \quad y_r = 3\left[\tanh\frac{t-9}{2} - \tanh\frac{t-19}{2}\right]. \tag{2.41}$$

The lane-change expression produces a lateral excursion of approximately 6 m and a return, rather than assuming a standard lane width. The oval changes both reference speed and curvature; the double-loop trajectory reverses curvature and crosses itself. Time-indexed tracking removes ambiguity at that crossing.

Each run begins with the same reference-frame error and previous command:

$$\boldsymbol{e}_0 = [-1\text{ m},\ 1.4\text{ m},\ 8^\circ,\ -0.7\text{ m/s}]^\top, \qquad \boldsymbol{u}_{-1} = 0. \tag{2.42}$$

Only the lane-change plant receives $d_y = 0.35$ m/s and $d_a = -0.6$ m/s^2 during $24 \le t < 26$ s. These values are not included in the controller's preview. Durations are 52, 64, and 36 s, producing 520, 640, and 360 optimizations respectively. Each trajectory includes the final state sample, for 1,523 state samples in total.

2.11.2 Metrics and sampling conventions

For a run containing M state samples,

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{M}\sum_{j=0}^{M-1}[(x_j - x_{r,j})^2 + (y_j - y_{r,j})^2]}, \tag{2.43}$$

$$\mathrm{RMSE}_y = \sqrt{\frac{1}{M}\sum_{j=0}^{M-1} e_{y,j}^2}, \qquad \mathrm{RMSE}_v = \sqrt{\frac{1}{M}\sum_{j=0}^{M-1} e_{v,j}^2}. \tag{2.44}$$

Full-run values include startup recovery and the disturbance. Orthogonality of the frame transformation gives $(x - x_r)^2 + (y - y_r)^2 = e_s^2 + e_y^2$. Position RMSE therefore includes along-track timing error; it is not interchangeable with cross-track RMSE.

The simulation output also contains cross-track RMSE restricted to $t \geq 5$ s. Those restricted values are discussed only as supplemental diagnostics and are not substituted for the primary full-run results. Control-rate checks include the transition from the zero initial command.

3 Results

3.1 Trajectory tracking results

Figure 2 compares the reference and simulated vehicle trajectories.

Scenario	Position RMSE [m]	Cross-track RMSE [m]	Speed RMSE [m/s]
Variable-speed oval	0.3180	0.2498	0.1206
Double-loop	0.2893	0.1966	0.1246
Disturbed lane change	0.3958	0.2708	0.1707

Table 3.1: Full-run tracking errors for the three test cases.

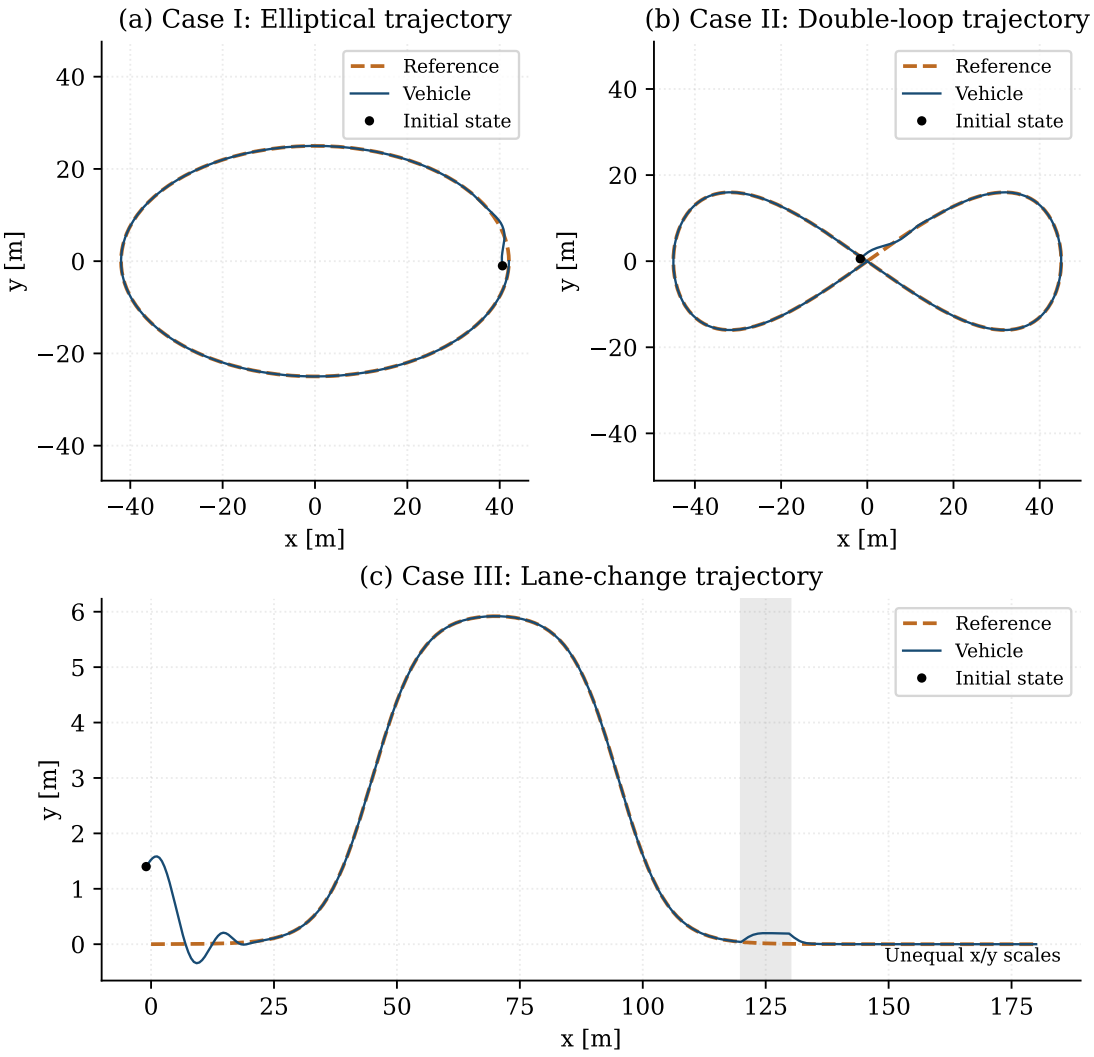


Figure 2: Reference and simulated trajectories for the three test cases.

Figure 2(a)–(c) shows convergence toward the reference after the initial offset for Cases I–III. The shaded region in panel (c) marks reference travel during the disturbance interval; its horizontal and vertical axes use different scales. Maximum position errors are 1.935, 2.038, and 2.034 m respectively. Those peaks exceed the initial position error $\sqrt{1^2 + 1.4^2} \simeq 1.720$ m, so the transient should not be described as monotonic error reduction. The initial heading error and command-rate limits matter during recovery.

3.2 Tracking errors for the three test cases

The error responses are presented separately for each trajectory. In each figure, panel (a) shows the signed cross-track error and panel (b) shows the speed error. Both quantities are measured relative to the reference motion at the same time.

3.2.1 Case I: Elliptical trajectory

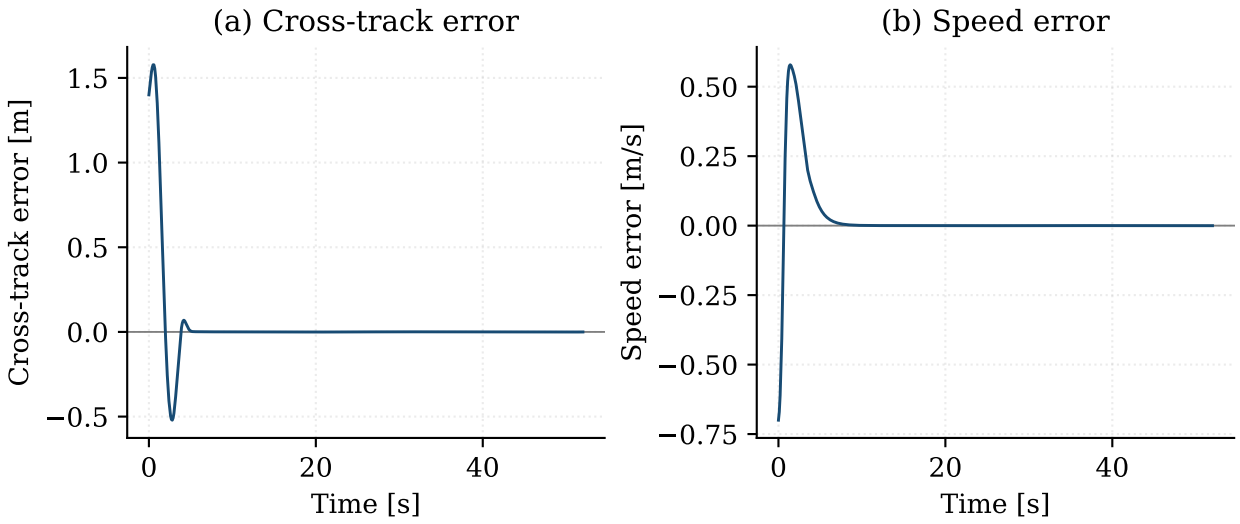


Figure 3: Tracking errors for the elliptical trajectory.

Figure 3(a) shows an initial cross-track transient caused by the position and heading offsets. The error changes sign before approaching zero. Figure 3(b) shows the speed error recovering from its initial negative value. After the first five seconds, cross-track RMSE is 0.000587 m, compared with a full-run value of 0.2498 m. This difference indicates the contribution of startup recovery to the full-run error.

3.2.2 Case II: Double-loop trajectory

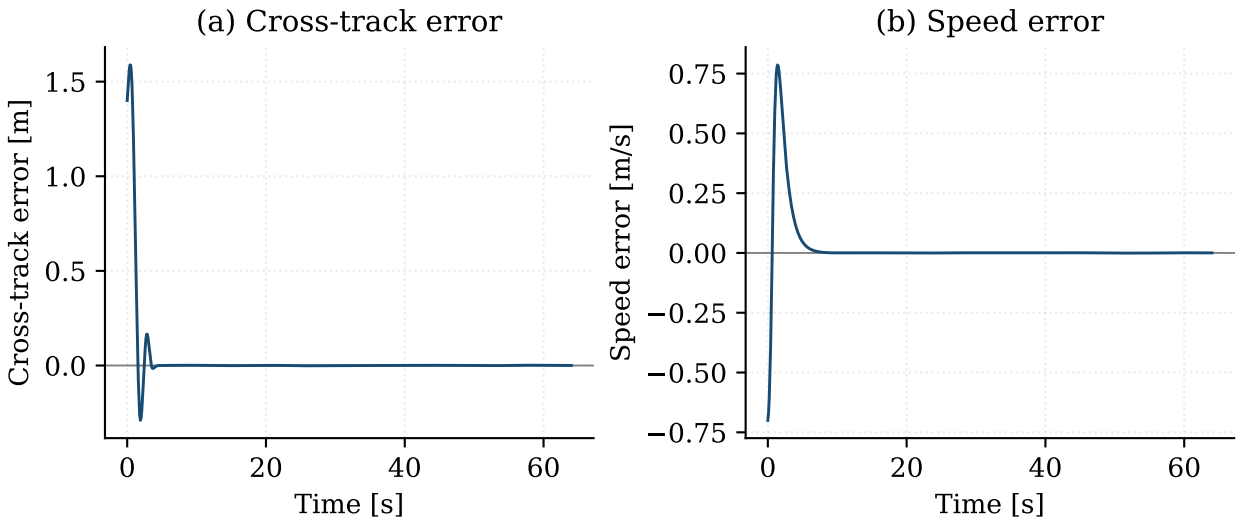


Figure 4: Tracking errors for the double-loop trajectory.

Figure 4 shows recovery from the initial cross-track and speed errors while the reference follows two connected loops. The controller tracks a time-indexed reference, so the central crossing does not require selecting between two geometric path branches. Cross-track RMSE after the first five seconds is 0.000935

m; the full-run value is 0.1966 m. These small post-startup errors are obtained with exact state feedback and a kinematic plant.

3.2.3 Case III: Disturbed lane-change trajectory

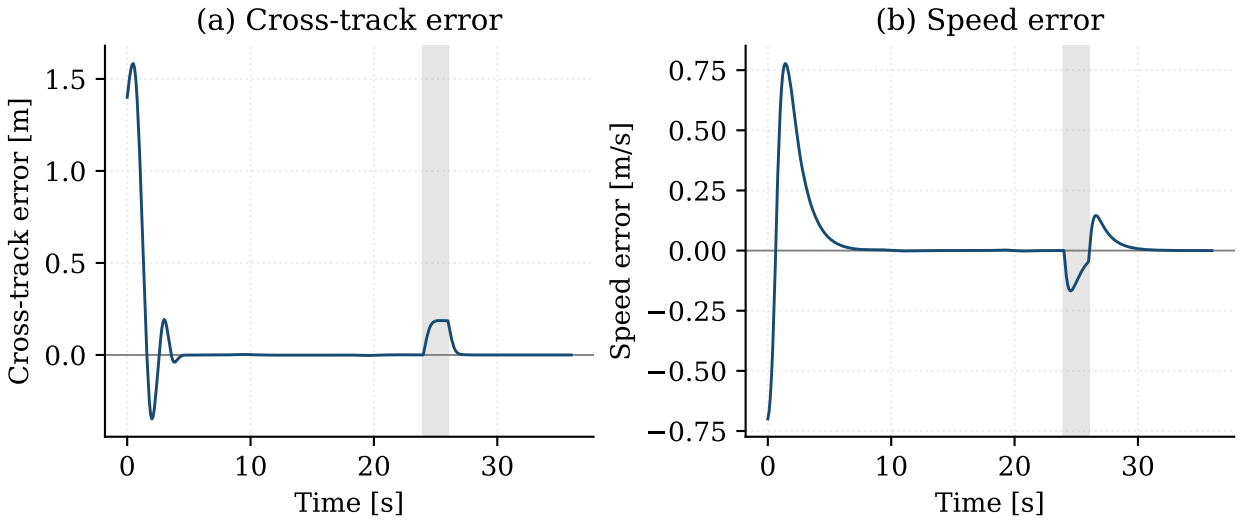


Figure 5: Tracking errors for the disturbed lane-change trajectory.

Figure 5 shows startup recovery followed by a second transient during the disturbance interval, shaded from 24 to 26 s. The lateral drift increases cross-track error, while the longitudinal load changes speed error. Both quantities return toward zero after the disturbance ends. Cross-track RMSE for $t \geq 5$ s is 0.0441 m, including the disturbance, compared with 0.2708 m over the full run.

3.3 Disturbance response and recovery analysis

Figure 6 shows the vehicle response during and after the disturbance. In Figure 6, gray shading marks 24–26 s, dotted orange lines show the recovery thresholds, and the green line marks sustained threshold satisfaction at 27.3 s. The temporary disturbance is injected into the nonlinear plant only. Positive global lateral drift initially moves the vehicle away from its reference, while negative longitudinal load reduces speed. The next optimization receives the resulting tracking error and adjusts steering and acceleration.

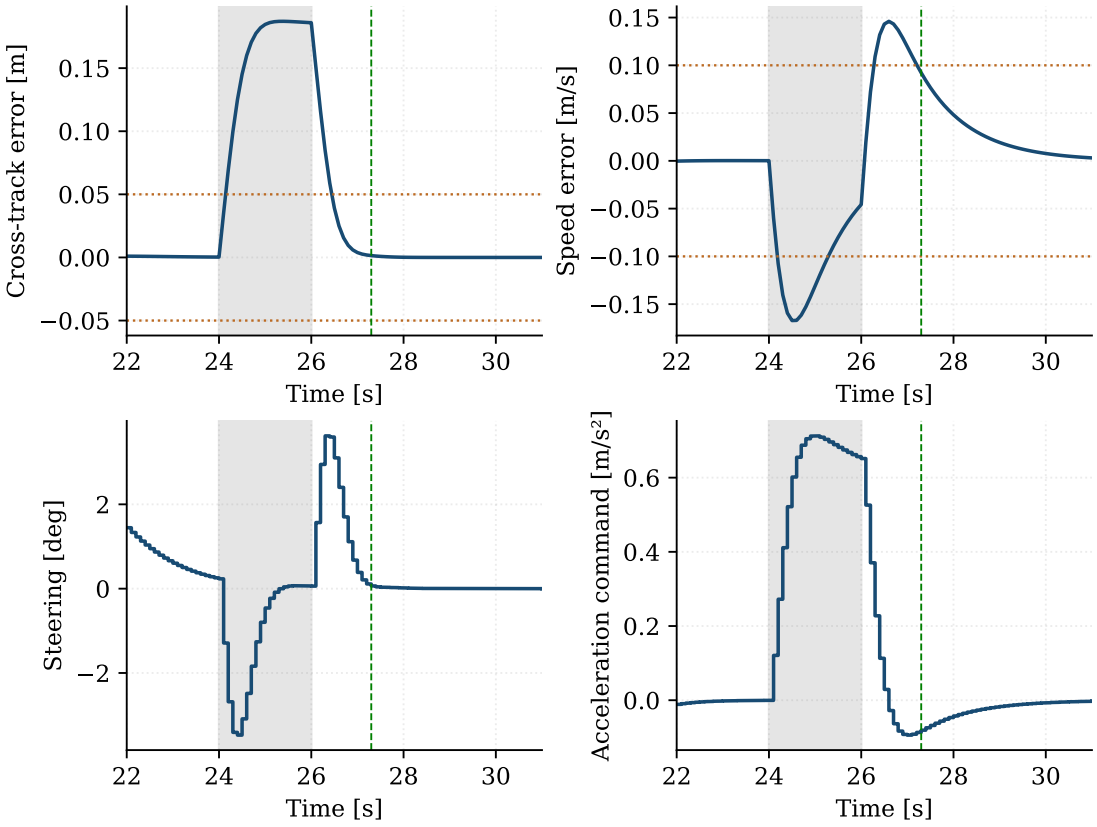


Figure 6: Lane-change tracking errors and control inputs during disturbance recovery.

Quantity	Value
Maximum $\lvert e_y \rvert$ during $24 \le t < 26$ s	0.1871 m
Maximum $\lvert e_v \rvert$ during $24 \le t < 26$ s	0.1672 m/s
Cross-track error at 26 s	0.1860 m
Speed error at 26 s	−0.0457 m/s
First sustained threshold satisfaction after removal	27.3 s
Delay from disturbance removal to that sample	1.3 s

Recovery is defined as the first sampled time at or after 26 s for which $\lvert e_y \rvert \le 0.05$ m and $\lvert e_v \rvert \le 0.10$ m/s remain satisfied through the end of the saved run. This is a post-processing criterion chosen for analysis, not a controller requirement. It does not test along-track or heading settling, and the 0.1 s sampling limits its time resolution. The result demonstrates finite-run recovery from this specific temporary disturbance, not a bound for all disturbances.

3.4 Command limits and optimization outcomes

Figure 7 compares the applied commands and their rates with the configured limits.

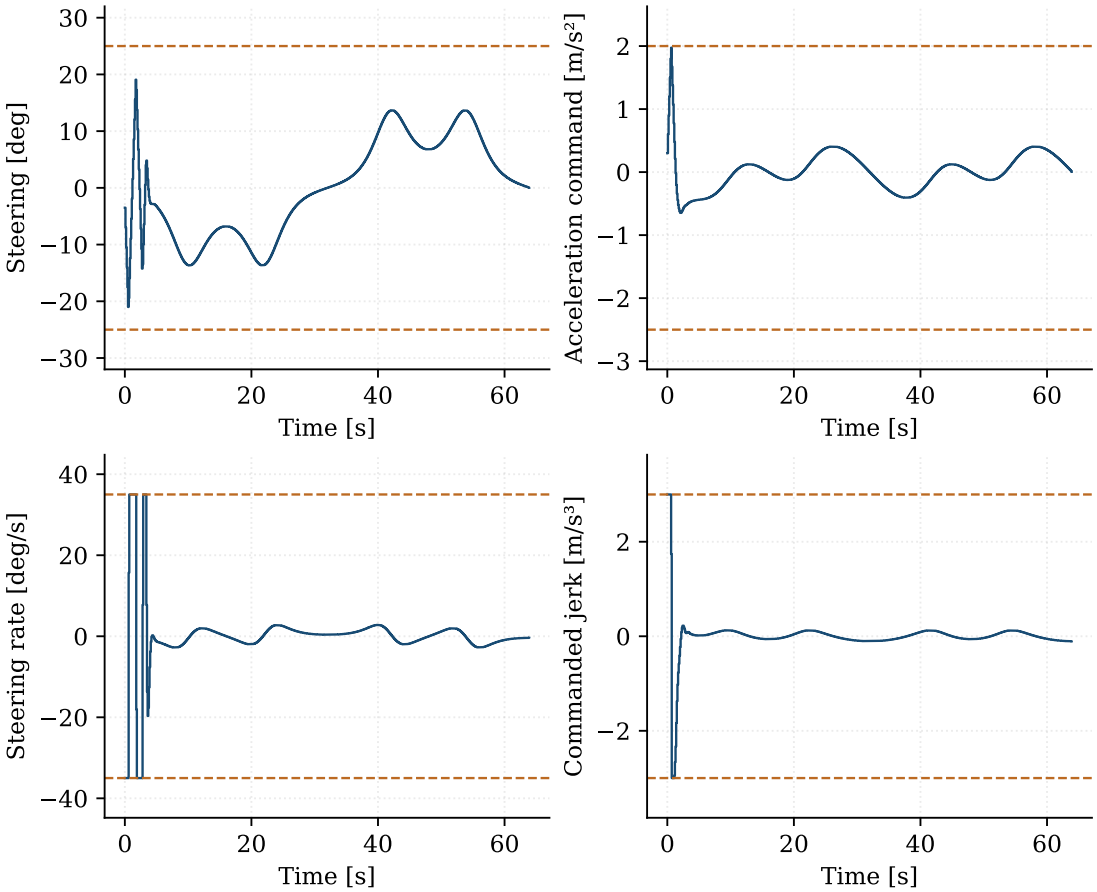


Figure 7: Steering, acceleration and command rates for Case II.

Recorded quantity	Oval	Double-loop	Lane change
Peak $\lvert\delta\rvert$ [deg]	25.000	21.000	21.265
Peak $\lvert\Delta\delta\rvert/T_s$ [deg/s]	35.000	35.000	35.000
Peak $\lvert\Delta a\rvert/T_s$ [m/s^3]	3.000	3.000	3.000
Acceleration range [m/s^2]	$[-0.248, 1.869]$	$[-0.647, 1.974]$	$[-0.417, 2.000]$
Speed range [m/s]	$[2.321, 5.075]$	$[2.710, 6.107]$	$[4.300, 5.778]$
Successful QP solves	520	640	360

In Figure 7, dashed lines indicate the configured limits. The command-rate traces reach these limits during parts of the maneuver, showing the influence of rate constraints on the correction. All evaluated command, rate and sampled-speed limits are satisfied within their configured tolerances. The largest recorded QP constraint residual is approximately 7.16×10^{-12} across the three runs. Tiny floating-point excursions around nominal limits are evaluated using the implementation's tolerances.

Recorded median controller times are approximately 0.899, 0.938, and 0.937 ms, with 95th-percentile times of 1.279, 1.075, and 1.403 ms. These stored local timings include QP assembly, setup, and solve, exclude plant integration, and lack a documented hardware specification. They are diagnostic measurements, not a transferable real-time benchmark or deadline guarantee.

4 Discussion

4.1 Tracking performance and recovery

The three experiments produced full-run position RMSE between 0.289 and 0.396 m while all 1,520 quadratic programs were solved successfully. These errors include recovery from the common initial offset and, for the lane-change run, the temporary disturbance. The results indicate that the reference-scheduled prediction model and receding-horizon feedback maintained tracking through changing speeds and curvature within the evaluated scenarios.

The lane-change experiment illustrates the role of repeated state feedback when disturbances are absent from the prediction model. After the disturbance ended, cross-track and speed errors satisfied the stated settling thresholds within 1.3 s. This value depends on the selected thresholds and the 0.1 s sampling interval; it characterizes recovery for this experiment rather than a general disturbance bound.

4.2 Constraints and command smoothness

The rate constraints shape the correction available at each control step. Saturated steering-rate and acceleration-rate traces show that recovery sometimes requires the full permitted command change. The optimizer then distributes further correction over the horizon. The tracking, feedforward and rate penalties balance error reduction against command magnitude and smoothness, while the hard constraints specify allowable motion and input changes.

4.3 Model assumptions and limitations

The plant assumes planar kinematic motion, exact state feedback and instantaneous command application. Tire slip, sensor noise, state-estimation error, actuator lag and communication delay are excluded. The results therefore establish performance for the stated simulation conditions; validation on a physical vehicle requires these additional effects to be modeled and tested.

Reference scheduling provides a sequence of local linear models along the known trajectory. Linearization and forward-Euler discretization introduce prediction error, which is corrected through feedback at subsequent samples. Since the formulation contains neither integral-error states nor a disturbance observer, zero steady-state error under a persistent unknown load is not guaranteed.

Each feasible quadratic program has a unique optimum under the stated positive-definite cost structure. This optimization property is distinct from closed-loop stability. The terminal weight encourages a small horizon-end error, but a recursive-feasibility or stability guarantee would require additional terminal conditions or an explicit invariant-set argument [1].

4.4 Future work

A matched comparison with fixed-model MPC would help isolate the benefit of reference scheduling. Further studies should vary horizon length and rate penalties, introduce persistent-disturbance estimation, and evaluate noisy or delayed measurements, actuator dynamics and tire-slip effects. These extensions would clarify the tracking–smoothness trade-off and the operating range of the controller.

References

[1] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed., Nob Hill Publishing. https://sites.engineering.ucsb.edu/~jbraw/mpc/.

[2] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: An Operator Splitting Solver for Quadratic Programs," *Mathematical Programming Computation*, vol. 12, no. 4, pp. 637–672, 2020.

[3] OSQP developers, "The solver," OSQP documentation. https://osqp.org/docs/solver/.